
PILLAR 07

The Bill — Infrastructure & AI Cost Model

Bottom-up unit economics across three scales.

Respect the Funk · marketing engine research program

Target: *Losing Sleep* — Hallow Youth · 15 July 2026

516 lines · 94 source links

Pillar 7 — Infrastructure & AI Cost Model ("The Bill")

Pricing verified as of 2026-07-15. Every price below links to the vendor page it was read from. Prices decay fast — re-verify anything older than ~60 days. Items that could not be confirmed against a vendor's own page are marked **UNVERIFIED** — **needs check** rather than guessed.

Bottom line

SCENARIO	FIXED (SUBS)	VARIABLE (USAGE)	
Scrappy — 1 song, 20 movies, 100 posts/mo, ~20 cutscenes	\$20	~\$35	≈ \$55/mo
Growth — 100 movies, 1,000 posts/mo, ~200 cutscenes	\$27	~\$335	≈ \$360/mo
Scale — 1,000 movies, 10k posts/mo, ~2,000 cutscenes	\$62	~\$3,350	≈ \$3,400/mo

Excludes ad spend (Pillar 2) — at Growth/Scale, ad spend dwarfs all infrastructure below.

The three things that matter:

- AI video generation is the bill.** Everything else — hosting, DBs, LLM tokens, indexing, storage — rounds to noise. At Growth, ~92% of variable cost is AI footage generation. At Scale, ~93%. Choosing Veo 3.1 Standard (\$0.40/s) over Veo 3.1 Lite (\$0.05/s) turns the Scale bill from **\$3,400** → **\$24,300/mo**. This single dropdown is worth more than every other optimization in this document combined.
- The fixed floor is ~\$20/mo and it's almost all one Vercel seat.** Clerk (50k MAU), PostHog (1M events), Neon and Upstash free tiers cover the entire webapp through 1,000 users. At Scrappy volume, fixed cost is ~36% of the bill despite the product barely being used — normal, and fine, because the floor is \$20 not \$500.
- Everything else is ~free.** Self-hosted scene detection is ~\$0.03/movie vs \$18 on Google Video Intelligence (600x). Subtitles beat transcription. pgvector beats every dedicated vector DB at our scale. LLM cost per post is \$0.006.

Assumptions (stated up front)

ASSUMPTION	VALUE	WHY
Movie proxy format	1080p H.264 @ 8 Mbps, 2 hr avg	→ 7.2 GB/movie
Generated clip	30s vertical 9:16 @ 10 Mbps	→ 37.5 MB/clip
Frame sampling	Keyframes only (~1,000 shots/movie)	7.2x cheaper than 1 fps; see §3
Embedding dim	768-dim float32 = 3,072 B/vector	→ 1M vectors ≈ 3 GB raw
Post generation	1,500 in + 300 out tokens	Brand voice + song context → copy
Cutscene reasoning	8,000 in + 500 out tokens	Scene candidates + params → EDL
Render job	90s active CPU, 3 GB mem	FFmpeg burn-in, 30s 1080p

ASSUMPTION	VALUE	WHY
Ad spend	Excluded	Owned by Pillar 2

Context note: this is now a **multi-tenant** engine for labels/artists on Meta + TikTok, and **scale-to-zero is a hard requirement**. Both shape the recommendations — see §12 and §13.

1. Video processing (FFmpeg render)

Unit: render one 30-second 1080p vertical video with audio + burned-in subtitles.

VENDOR	RATE	MATH	
Vercel Fluid Compute	\$0.128/Active-CPU-hr + \$0.0106/GB-hr	(90/3600)×0.128 + (3×100/3600)×0.0106	\$0.0041
AWS Lambda	\$0.0000166667/GB-s	3 GB × 90 s = 270 GB-s	\$0.0045
Coconut.co	\$0.015/min HD	0.5 min × 0.015	\$0.0075
AWS MediaConvert (Professional — Basic tier <i>cannot</i> do caption burn-in)	\$0.012/min base, 2x HD	0.5 × 0.024	\$0.012+
Mux	encoding/storage/delivery	—	N/A — ABR streaming pipeline, not an FFmpeg job runner
Cloudflare Stream	free encode, \$0.005/min storage	—	N/A — same shape mismatch
Transloadit	\$69/mo, \$1.80/GB overage	billed on GB in+out	\$0.05–\$3.60 ⚠️ blows up if the job imports a full source movie

Verdict: Vercel Fluid or Lambda (~\$0.004/render). Both ~3x cheaper than Coconut, ~3–6x cheaper than MediaConvert. Pick whichever the app already lives on.

Serverless limits are a real constraint

LIMIT	VERCEL	LAMBDA
Max duration	Hobby 300s hard; Pro 800s GA / 1800s beta	900s hard cap — default is 3s, must be raised
Max memory	Hobby 2 GB / 1 vCPU; Pro 4 GB / 2 vCPU	10,240 MB
Ephemeral disk	—	512 MB default; extra billed \$0.0000000309/GB-s — a 7.2 GB movie will not fit

The ephemeral-disk ceiling is the real trap: you cannot pull a whole movie into a Lambda. Architecture must stream byte-ranges of the source, not download it. (Architecture is another agent's call — flagging the cost consequence only.)

2. Scene detection & indexing — the 600x decision

Unit: index one 2-hour (120 min) movie.

VENDOR	RATE	
PySceneDetect (self-hosted)	~30 min CPU @ \$0.0472/hr Modal / ~\$0.0416/hr t3.medium ¹	~\$0.02–0.04
TwelveLabs	\$0.0435/min incl. infra	\$5.22
Google Video Intelligence — shot only	\$0.05/min	\$6.00
AWS Rekognition Video — segment only	\$0.05/min	\$6.00
Azure Video Indexer — Standard	\$0.09/min ²	\$10.80
Google / AWS / Azure — shot + label (full)	\$0.15/min	\$18.00

¹ AWS EC2's own pricing page is JS-rendered; figure cross-checked via Vantage mirror. Processing time is an engineering estimate (~60–150 fps decode), not a vendor SLA. ² Azure's pricing page is JS-rendered; figure pulled from Microsoft's own public Retail Prices API (primary source).

Scaling

VENDOR	10 MOVIES	100	1,000
PySceneDetect	\$0.30	\$3	\$30
TwelveLabs	\$52	\$522	\$5,220
Google/AWS (shot+label)	\$180	\$1,800	\$18,000
Azure Standard	\$108	\$1,080	\$10,800

Verdict: PySceneDetect, unambiguously. 600x cheaper than Google full-indexing at 1,000 movies (\$30 vs \$18,000). The catch: it gives shot boundaries only, so you run your own labels/embeddings on top — which \$3 shows costs ~\$3 for the whole 1,000-movie library. Total DIY: **~\$33 vs \$18,000.** There is no scale at which the managed APIs win here.

3. Embeddings & vector search

Sampling assumption dominates everything

2-hour movie = 7,200 s. CLIP ViT-B/32 at ~100 frames/s sustained on an A10 (engineering estimate).

STRATEGY	FRAMES	SELF-HOSTED (MODAL A10 \$1.10/HR)	REPLICATE (\$0.00036/RUN)
1 fps	7,200	\$0.022	\$2.59
1 frame / 5s	1,440	\$0.0044	\$0.518
Keyframes only (~1,000)	1,000	\$0.0031	\$0.36

7.2x swing from the sampling choice alone. On self-hosted GPU it's noise either way (\$0.02 vs \$0.003); on per-run-billed Replicate it's real money at 1,000 movies: **\$2,590 (1 fps) vs \$360 (keyframes)**. Use keyframes — scene detection already gives you the shot list for free.

Vector DB at 100k / 1M / 10M vectors

VENDOR	100K	1M	10M	SCALE-TO-ZERO?
pgvector on Neon (\$0.35/GB-mo)	~\$0.21	~\$2.10	~\$21	✅ autosuspend
pgvector on Supabase (8 GB incl. in Pro)	\$0 marginal	\$0 marginal	~\$6.50 marginal	❌ \$25/mo floor
Turbopuffer (\$0.02/GB-mo)	\$16 (min)	\$16 (min)	\$16 (min)	❌ \$16/mo min
Pinecone	\$20 (Builder)	~\$50 (Standard min)	\$50–105	❌ \$20/mo min
Weaviate Cloud (Flex)	\$45 (min)	\$45 (min)	\$45 (min)	❌ \$45/mo min
Qdrant Cloud	~\$51	~\$512	~\$5,125	❌ RAM-resident
LanceDB	UNVERIFIED — Cloud in free beta, no published pricing		—	—

Qdrant's ~\$0.078/GB-RAM-hr is third-party-derived (their calculator is JS-only) — **UNVERIFIED against a primary source**. Quantization/on-disk mode would cut it 4–32x.

Is a vector DB even needed? No.

1,000 movies × ~1,000 shots = **~1M vectors**. pgvector HNSW handles that with sub-100ms ANN latency for an interactive workload. We need Postgres anyway for tenants/movies/scenes/licensing. Colocating means filtered search ("similar shots AND licensed AND tenant=X") is one SQL query, not a cross-service join. Marginal cost ~\$2/mo vs \$16–\$512/mo for a dedicated service — every one of which has a **monthly minimum that violates the scale-to-zero requirement**.

Use **pgvector on Neon**. Revisit only above ~10M vectors or if QPS outgrows a shared instance.

4. Transcription — probably \$0

VENDOR	RATE	
Existing subtitle files	OpenSubtitles / bundled	~\$0
Deepgram Nova-3 (pre-recorded)	\$0.0077/min	\$0.46 (\$200 free credit)
Deepgram Nova-3 (streaming)	\$0.0048/min	\$0.29
Groq / OpenAI Whisper / AssemblyAI	—	UNVERIFIED — needs check
Self-hosted Whisper	\$0.59/hr T4 ÷ ~20x realtime	~\$0.03

Movies already have subtitles. For product (B)'s library, dialogue is a solved, near-free problem — a `.srt` gives you exact text *and* timecodes, which is precisely what "dialogue over the cut" needs. Even at full price, transcribing 1,000 movies (2,000 hrs) via Deepgram is a **one-time ~\$920**, not recurring. This line item is a rounding error either way; treat as **\$0 (subtitles)** in the bill.

⚠️ **The pivot changes this.** Moving to AI-generated / stock footage means there are no pre-existing subtitles for generated clips. Dialogue must then come from the generation prompt or TTS — cost unmodeled here. **Gap flagged.**

5. LLM costs

All rates from Anthropic's pricing docs, verified 2026-07-15:

MODEL	INPUT / MTOK	OUTPUT / MTOK	CACHE WRITE (5M)	CACHE READ	BATCH IN/ OUT
Claude Opus 4.8	\$5	\$25	\$6.25	\$0.50	\$2.50 / \$12.50
Claude Sonnet 5 (intro, thru Aug 31 2026)	\$2	\$10	\$2.50	\$0.20	\$1 / \$5
Claude Sonnet 5 (from Sep 1 2026)	\$3	\$15	\$3.75	\$0.30	\$1.50 / \$7.50
Claude Haiku 4.5 (<code>claude-haiku-4-5-20251001</code>)	\$1	\$5	\$1.25	\$0.10	\$0.50 / \$2.50

⚠️ **Sonnet 5 introductory pricing ends Aug 31, 2026** — a +50% price increase lands in ~6 weeks. Model the bill at \$3/\$15. ⚠️ Opus 4.7+ / Sonnet 5 use a **newer tokenizer producing ~30% more tokens for the same text** — real cost is ~30% above naive estimates on those models.

Levers: Batch API = **flat 50% off in and out**. Cache read = **0.1x input** (write 1.25x for 5m / 2x for 1h) — pays for itself after **one** read at 5m. **Discounts stack.**

Competitors ([Gemini](#)): Gemini 3.1 Flash-Lite \$0.25/\$1.50; Gemini 2.5 Flash \$0.30/\$2.50; Gemini 3.1 Pro \$2/\$12. OpenAI: **UNVERIFIED — needs check.**

Cost per generated post (1,500 in + 300 out)

CONFIG	MATH	
Sonnet 5, naive	$(1500 \times \$2 + 300 \times \$10) / 1M$	\$0.0060
Sonnet 5 + cache (1,200 cached)	$(300 \times \$2 + 1200 \times \$0.20 + 300 \times \$10) / 1M$	\$0.0038 (−36%)
Sonnet 5 + cache + batch	above $\times 0.5$	\$0.0019 (−68%)
Haiku 4.5, naive	$(1500 \times \$1 + 300 \times \$5) / 1M$	\$0.0030
+ render (§1)	+\$0.0041	≈ \$0.008–0.010/post

10,000 posts/mo ≈ \$60 of LLM + \$41 of render ≈ \$100. LLM tokens are not the bill.

Cost per generated cutscene (8,000 in + 500 out)

CONFIG	
Sonnet 5	\$0.021
Opus 4.8	\$0.053
Sonnet 5 + batch	\$0.011
1,000 overnight cutscene-selection calls, Sonnet 5 batch	\$10.50 (vs \$21 sync)

The reasoning call is ~0.7% of a cutscene's cost. Footage is ~99%. Do not optimize the LLM here — use Opus if it picks better scenes; the \$0.03 delta is irrelevant against \$1.50+ of footage.

6. AI video generation — now the primary cost driver

Since the product pivoted from copyrighted film to public-domain / licensed stock / AI-generated footage, this is no longer an alternative — it's the main input.

Unit: 30 seconds of AI video.

VENDOR / MODEL	RATE		SOURCE
Veo 3.1 Lite (720p)	\$0.05/s	\$1.50	ai.google.dev
Veo 3.1 Lite (1080p)	\$0.08/s	\$2.40	same
Veo 3.1 Fast (720p)	\$0.10/s	\$3.00	same
Veo 3.1 Fast (1080p)	\$0.12/s	\$3.60	same
Veo 3.1 Standard (720p/1080p)	\$0.40/s	\$12.00	same
Veo 3.1 Standard (4K)	\$0.60/s	\$18.00	same
Runway Gen-4 Turbo	5 cr/s @ \$0.01/cr = \$0.05/s	\$1.50	docs.dev.runwayml.com
Runway Gen-4.5	12 cr/s = \$0.12/s	\$3.60	same
Runway Seedance 2.0 (1080p)	40 cr/s = \$0.40/s	\$12.00	same
Runway Seedance 2.0 (4K)	150 cr/s = \$1.50/s	\$45.00	same
Kling 3.0	~\$0.10/s	~\$3.00	UNVERIFIED — third-party only
Luma Ray 2	~\$0.04/s	~\$1.20	UNVERIFIED — third-party only
Sora / Pika	—	—	UNVERIFIED — needs check

⚠️ Veo generates max 8s per call → 30s = 4 generations (32s billed). Costs above already reflect per-second billing; add ~7% for the rounding.

Images (Imagen 4): Fast \$0.02, Standard \$0.04, Ultra \$0.06/image. OpenAI/Midjourney/Stability: UNVERIFIED — needs check.

The footage decision, priced

FOOTAGE SOURCE	COST/30S CUTSCENE	SCALE (2,000/MO)
Licensed stock / public domain (library amortized)	~\$0.03	~\$60
Veo 3.1 Lite / Runway Gen-4 Turbo	~\$1.55	~\$3,100
Veo 3.1 Fast 1080p / Runway Gen-4.5	~\$3.65	~\$7,300
Veo 3.1 Standard	~\$12.05	~\$24,100
Seedance 2.0 4K	~\$45	~\$90,000

This table is the whole cost model. A 400x spread on one dropdown. Public-domain/stock footage is ~50x cheaper than even the cheapest AI generation — the pivot's legal motivation and its cost motivation point the same direction, which is a rare gift. Default to stock/PD; use Veo Lite/Gen-4 Turbo for gap-filling; gate Veo Standard behind an explicit per-tenant budget flag.

7. Storage & bandwidth

VENDOR	STORAGE /GB-MO	EGRESS	FREE TIER
Cloudflare R2	\$0.015	\$0 — confirmed, uncapped	10 GB, 1M/10M ops
Backblaze B2	\$0.00695	3x storage free, then \$0.01/GB; free via Bandwidth Alliance CDN	10 GB
AWS S3	\$0.023	100 GB free, then \$0.09/GB	—
Vercel Blob	\$0.023	\$0.05/GB	1 GB / 10 GB transfer

100-movie library (720 GB) + 1,000 clips (37.5 GB) = 758 GB

VENDOR	STORAGE	EGRESS (11.1 TB/MO ¹)	
Cloudflare R2	\$11.37	\$0	\$11.37
B2 + Cloudflare (Bandwidth Alliance)	\$5.27	\$0	\$5.27
B2 standalone	\$5.27	\$88.26	\$93.53
Vercel Blob	\$17.43	\$550.00	\$567.43
AWS S3	\$17.43	\$985.00	\$1,002.43

¹ Assumes 100 movies re-pulled 5x/mo (3.6 TB) + 1,000 clips × 200 views × 37.5 MB (7.5 TB).

Egress is the hidden killer

S3 costs 88x more than R2 for the identical bytes (\$1,002 vs \$11). S3's egress bill (\$985) is 56x its own storage bill (\$17). Storage is never the problem; movement is. Use R2 (or B2+Cloudflare if squeezing). Note this is a *per-tenant-linear* cost — 10 tenants × 100 movies = \$114/mo on R2, or **\$10,000/mo on S3**.

⚠️ **Storage does not scale to zero.** A 1,000-movie library is ~\$110/mo on R2 whether or not anyone logs in. This is the only meaningful always-on cost besides the Vercel seat.

8. App hosting & data (Product A)

VENDOR	FREE TIER	PAID	SCALE-TO-ZERO?
Vercel	Hobby: 1M reqs, 100 GB, 4 Active-CPU-hrs (non-commercial only)	\$20/seat/mo + \$0.128/Active-CPU-hr, \$0.0106/GB-hr, \$0.60/1M invocations	✗ \$20 seat is always-on
Neon	0.5 GB, 100 CU-hr	\$0.106/CU-hr (Launch), \$0.35/GB-mo, no base fee	✓ autosuspend, no minimum
Supabase	500 MB, 50k MAU	\$25/mo Pro flat	✗ \$25 floor
Upstash Redis	256 MB, 500k cmds/mo	\$0.20/100k cmds PAYG	✓ true PAYG
Clerk	50,000 MAU	\$0.02/MAU beyond	✓ free tier covers us entirely

Assumptions: ~600 API reqs/user/mo, ~50ms Active CPU/req, media in blob storage not Postgres.

LINE	10 USERS	100 USERS	1,000 USERS
Vercel	\$20 (seat; usage ~\$0)	\$20	\$20 + ~\$1.70 = \$22
Neon	\$0 (free tier)	~\$5.12 (1 GB storage breaks 0.5 GB free)	~\$21.55
Upstash	\$0 (9k cmds)	~\$1.80 (900k cmds)	~\$18.00
Clerk	\$0	\$0	\$0
Total	\$20	~\$27	~\$62

Findings: (a) Clerk's 50k-MAU free tier covers all three scales — auth is \$0 until we're 50x bigger. (b) Vercel *usage* is ~\$1.70/mo even at 1,000 users; the **\$20 seat, not scale, is the entire Vercel bill** — and it's a commercial-ToS requirement, not a usage cost. (c) DB **storage size**, not compute, forces the first paid upgrade. (d) **Choose Neon over Supabase:** Neon has no base fee and autosuspends → satisfies scale-to-zero; Supabase's \$25 flat floor does not.

⚠️ Vercel Pro's "usage credit" amount is referenced in docs but **not quantified on any official page — UNVERIFIED.**

9. Analytics

VENDOR	FREE TIER	PAID
PostHog	1M events/mo, 5k replays	\$0.00005/event (1–2M band)
Mixpanel	1M events/mo	\$0.28/1k events
Plausible	none (30-day trial)	\$9–19/mo @ 10k pageviews
Dub.co	none	\$90/mo Business floor
Bitly	5 links/mo	\$10 Core / \$29 Growth / \$199 Premium
Self-built smart link (Vercel Edge + Postgres)	—	~\$0–15/mo marginal

At ~100 events/user/mo:

VENDOR	10 USERS (1K EV)	100 (10K EV)	1,000 (100K EV)
PostHog	\$0	\$0	\$0
Mixpanel	\$0	\$0	\$0
Plausible	\$9–19	~\$14–39	UNVERIFIED
Self-built links	~\$0	~\$0–2	~\$5–15

Verdict: PostHog free + self-built smart links = \$0. PostHog's 1M-event ceiling isn't hit until ~10,000 users. Plausible is the *only* option that costs money at 10 users. Dub's \$90/mo floor to track 200 clicks is absurd at our scale — a redirect + a Postgres insert on infra we already pay for is ~free.

⚠️ Dub.co's Free/Pro tiers cited by 2026 third-party sites **do not appear on the live pricing page** — treat \$90 as the confirmed floor; cheaper tiers **UNVERIFIED / possibly discontinued**. Plausible's >10k-pageview tiers are unpublished — **UNVERIFIED**.

10. The bill

Scrappy — 1 song, 20 movies, 100 posts/mo, ~20 cutscenes/mo

TYPE	LINE	MATH	COST
FIXED	Vercel Pro seat	1 × \$20	\$20.00
FIXED	Neon	free tier	\$0
FIXED	Upstash	free tier	\$0
FIXED	Clerk	10 MAU < 50k	\$0
FIXED	PostHog	1k ev < 1M	\$0
	Fixed subtotal		\$20.00

TYPE	LINE	MATH	COST
VAR	Index 20 movies (PySceneDetect)	$20 \times \$0.03$	\$0.60
VAR	Embed keyframes	$20 \times \$0.003$	\$0.06
VAR	Transcription	subtitles	\$0
VAR	100 posts	$100 \times (\$0.006 + \$0.0041)$	\$1.01
VAR	20 cutscenes (Veo Lite)	$20 \times \$1.55$	\$31.00
VAR	Storage (144 GB, R2)	$144 \times \$0.015$	\$2.16
Variable subtotal			\$34.83
TOTAL			≈ \$55/mo

With stock/PD footage instead: variable drops to **\$3.83** → **total ≈ \$24/mo**.

Growth — 100 movies, 1,000 posts/mo, ~200 cutscenes/mo

TYPE	LINE	MATH	COST
FIXED	Vercel + Neon + Upstash + Clerk + PostHog	$\$20 + \$5.12 + \$1.80 + \$0 + \$0$	\$26.92
VAR	Index 100 movies	$100 \times \$0.03$	\$3.00
VAR	Embed keyframes	$100 \times \$0.003$	\$0.30
VAR	1,000 posts	$1,000 \times \$0.0101$	\$10.10
VAR	200 cutscenes (Veo Lite)	$200 \times \$1.55$	\$310.00
VAR	Cutscene reasoning	$200 \times \$0.021$	\$4.20
VAR	Storage (758 GB, R2)		\$11.37
VAR	Egress (11 TB, R2)		\$0
Variable subtotal			\$338.97
TOTAL			≈ \$366/mo

Veo Standard instead: **\$2,410 cutscenes** → **total ≈ \$2,466/mo**. Stock footage: **≈ \$56/mo**.

Scale — 1,000 movies, 10k posts/mo, ~2,000 cutscenes/mo

TYPE	LINE	MATH	COST
FIXED	Vercel + Neon + Upstash + Clerk + PostHog	$\$22 + \$21.55 + \$18 + \$0 + \$0$	\$61.55
VAR	Index 1,000 movies	$1,000 \times \$0.03$	\$30.00
VAR	Embed keyframes	$1,000 \times \$0.003$	\$3.00
VAR	pgvector (1M vectors)		\$2.10

TYPE	LINE	MATH	COST
VAR	10,000 posts (batch+cache)	10,000 × \$0.0060	\$60.00
VAR	2,000 cutscenes (Veo Lite)	2,000 × \$1.55	\$3,100.00
VAR	Cutscene reasoning (batch)	2,000 × \$0.011	\$22.00
VAR	Storage (7.3 TB, R2)	7,300 × \$0.015	\$109.50
VAR	Egress (R2)		\$0
	Variable subtotal		\$3,326.60
	TOTAL		≈ \$3,388/mo

Veo Standard instead: **\$24,100 cutscenes** → **total ≈ \$24,400/mo**. Stock footage: **≈ \$350/mo**.

Unit economics

METRIC	SCRAPPY	GROWTH	SCALE
Cost per post (variable only)	\$0.0101	\$0.0101	\$0.0060
Cost per post (fully loaded)	\$0.55	\$0.37	\$0.34
Cost per cutscene (Veo Lite)	\$1.57	\$1.57	\$1.56
Cost per cutscene (stock/PD)	\$0.03	\$0.03	\$0.03
Cost per cutscene (Veo Standard)	\$12.07	\$12.07	\$12.06
Cost per managed account/mo ¹	\$55	\$37	\$34
Fixed as % of bill	36%	7%	2%

¹ Assuming 1 / 10 / 100 tenant accounts respectively.

Read: variable unit costs are **flat** — this business has essentially no economies of scale on the marginal cutscene, because the AI-video vendor charges linearly per second. The only lever that moves the unit cost is *which footage source you pick*, not how much you buy. Fixed costs amortize nicely (36% → 2%), but they were never the problem.

11. What would actually blow up the bill

Ranked by expected damage:

- Defaulting to premium AI video.** Veo 3.1 Standard vs Lite at Scale: **\$3,400 → \$24,400/mo (7x)**. Seedance 2.0 4K: **~\$90,000/mo**. One dropdown. **Hard-gate premium tiers behind a per-tenant budget cap.**
- Users regenerating cutscenes.** Every "try again" is a **full \$1.55–\$12 re-spend** — footage generation is not cacheable across prompts. 5 retries/cutscene silently 5x's the entire bill. **Cap retries; charge for them; make previews cheap (Lite) and finals expensive (Standard).**
- Putting media on S3 instead of R2.** 88x on the same bytes. At 10 tenants: **\$114/mo → \$10,000/mo**. Pure egress.

4. **Using managed video-intelligence APIs.** Google full-indexing at 1,000 movies = **\$18,000** vs **\$30** self-hosted (600x). A single "let's just use the API for now" decision.
5. **Embedding at 1 fps on a per-run-billed API.** Replicate at 1 fps × 1,000 movies = **\$2,590** vs \$360 keyframes vs **\$3** self-hosted. 860x between worst and best.
6. **Sonnet 5's price increase on Sep 1, 2026 (+50%) plus the ~30% tokenizer inflation** on Opus 4.7+/Sonnet 5 — a silent ~2x on LLM lines if modeled off old numbers. (Still small in absolute terms.)
7. **A dedicated vector DB.** \$16–\$512/mo of pure new spend for something pgvector does for \$2 — and every one has a **monthly minimum that breaks scale-to-zero**. Qdrant at 10M vectors unquantized: **~\$5,125/mo**.
8. **Transloadit importing full movies.** Billed per GB in+out → **\$3.60 per render** instead of \$0.004 (900x) if the job pulls the source file.
9. **Runaway render loops / no timeout.** Lambda at 3s default fails; Lambda at 900s max on a stuck job burns 10 GB × 900s repeatedly. **Set explicit timeouts and a dead-letter queue.**
10. **Per-tenant fixed costs.** If each tenant needs its own Neon project / Pinecone index / Vercel seat, the \$20 floor becomes \$20 × N. **Multi-tenancy must share infra by row, not by instance** (see §12).
11. **Dub.co-style SaaS floors.** \$90/mo to track 200 clicks. Death by a dozen \$20–90/mo subscriptions is how the *fixed* side quietly becomes \$500/mo.

12. Multi-tenant: what amortizes vs what scales linearly

COST	BEHAVIOR	NOTE
Vercel seat (\$20)	Amortizes	One team, N tenants — unless per-tenant seats
Clerk (50k MAU free)	Amortizes	Huge headroom; free to ~50k end users total
PostHog (1M events)	Amortizes	~10k users before first dollar
Neon compute	Mostly amortizes	Shared instance, row-level tenant isolation
Postgres/vector storage	Linear	~\$2/mo per 1M vectors per tenant
Movie/asset storage	Linear	~\$110/mo per 1,000-movie library — the always-on floor
AI footage generation	Strictly linear	~\$1.55/cutscene, no volume discount found
Scene indexing	Linear but trivial	\$0.03/movie
LLM tokens	Linear , batch-discountable	\$0.006/post

Implication: the fixed floor stays ~\$20–60 no matter how many tenants — the engine genuinely is reusable. But ****~93% of the bill is strictly per-tenant-linear AI footage****, which means **infra cost must be priced into per-tenant billing, not absorbed**. At \$1.55/cutscene and 200 cutscenes/mo, a tenant costs ~\$310/mo in footage alone. Any pricing below that loses money on every account, at any scale.

13. Scale-to-zero audit (hard requirement)

COMPONENT	TRUE IDLE FLOOR	VERDICT
Vercel Pro seat	\$20/mo	✗ Unavoidable (commercial ToS) — this is the floor
R2 storage	~\$11/mo per 100 movies; ~\$110 per 1,000	✗ Bytes at rest cost money regardless of traffic
Neon	\$0	✓ Autosuspend, no base fee, usage-only
Upstash Redis	\$0	✓ True PAYG
Clerk	\$0	✓ Free to 50k MAU
PostHog	\$0	✓ Free to 1M events
Vercel Functions	\$0	✓ No invocations = no charge
Lambda / Modal / Replicate	\$0	✓ Per-second billing
Veo / Runway	\$0	✓ Pure per-use
Pinecone / Turbopuffer / Weaviate	\$16–\$45/mo	✗ Minimums — reject
Supabase Pro	\$25/mo	✗ Flat floor — prefer Neon
Dub.co	\$90/mo	✗ Reject — self-build
Qdrant	RAM-resident	✗ Reject

True idle floor: \$20/mo (Vercel seat) + storage. An idle Scrappy tenant costs ~\$22/mo; an idle 1,000-movie Scale library costs ~\$130/mo. Everything else genuinely goes to zero — *provided* we pick Neon over Supabase, pgvector over a vector DB, and self-built links over Dub. Those three choices are the difference between a \$20 floor and a \$150+ floor.

14. Cost-control levers, with breakevens

LEVER	SAVING	BREAKEVEN / NOTE
Stock/PD footage over AI gen	~50x per cutscene (\$0.03 vs \$1.55)	Immediate. Biggest lever in the document.
Veo Lite over Veo Standard	8x (\$1.50 vs \$12/30s)	Immediate; use Lite for previews, Standard only for finals
Self-host scene detection	600x (\$30 vs \$18,000 @ 1k movies)	Breakeven at ~2 movies (\$0.06 DIY vs \$36 API). Self-host from day one.
Self-host CLIP over Replicate	~120x (\$3 vs \$360 @ 1k movies)	Modal A10 \$1.10/hr ÷ Replicate \$0.00036/run → breakeven ≈ 3,000 frames ≈ 3 movies
Keyframes over 1 fps	7.2x on embeddings	Free — scene detection already yields the shot list

LEVER	SAVING	BREAKEVEN / NOTE
R2 over S3	88x on the media bill	Immediate, no breakeven
Subtitles over transcription	~\$920 one-time @ 1k movies	Immediate (but see §4 pivot caveat)
pgvector over Pinecone	\$2 vs \$50/mo	Immediate; also preserves scale-to-zero
Prompt caching	–36% on posts	Pays back after 1 cache read (5m write = 1.25x, read = 0.1x)
Batch API	–50% flat, stacks with caching	Free if latency-tolerant. Cutscene <i>selection</i> is inherently batchable overnight.
Haiku 4.5 routing	–50% vs Sonnet 5	Use for templating/classification; keep Sonnet/Opus for scene reasoning where quality→footage-spend leverage is 70:1
Precompute + cache footage	Up to 100% of a regeneration	A generated 30s clip costs \$1.55 to make and \$0.0006/mo to store on R2. Never regenerate anything you could have stored — storage is ~2,600x cheaper than regeneration.

The single highest-leverage line in this document: storing a generated clip forever costs less than 0.04% of generating it once. Cache aggressively, permanently, and at every level.

Known gaps

GAP	STATUS
OpenAI pricing (GPT models, Whisper API, image gen)	UNVERIFIED — needs check
Sora / Pika per-second pricing	UNVERIFIED — needs check. Given the pivot, worth closing.
Kling / Luma pricing	Third-party only (~\$0.10/s, ~\$0.04/s) — needs vendor confirmation. Luma at ~\$0.04/s would beat Veo Lite.
Groq / AssemblyAI transcription	UNVERIFIED — low priority (subtitles win)
Licensed stock footage libraries	NOT PRICED — now load-bearing post-pivot. The "\$0.03/cutscene" stock figure assumes an amortized library whose acquisition cost is unmodeled. Highest-value gap to close.
TTS / dialogue for AI-generated clips	NOT PRICED — the pivot removes pre-existing subtitles (§4)
Vercel Pro usage credit amount	UNVERIFIED — not published
Plausible >10k pageviews	UNVERIFIED — not published
Dub.co Free/Pro tiers	UNVERIFIED — absent from live page
Qdrant \$/GB-RAM-hr	UNVERIFIED — calculator-only
Mux 1080p encode rate	UNVERIFIED — 720p baseline only
Coconut.co subtitle burn-in support	UNVERIFIED — pricing confirmed, capability not

GAP	STATUS
MediaConvert burn-in multiplier	UNVERIFIED — beyond the 2x HD multiplier
AWS EC2 / Azure list prices	Via Vantage mirror / Azure Retail Prices API (JS-rendered official pages)
PySceneDetect throughput, CLIP fps	Engineering estimates, not vendor SLAs

Sources

All verified 2026-07-15.

Compute & video processing

- [Vercel Functions usage & pricing](#) · [Vercel Functions limitations](#) · [Vercel pricing](#) · [Vercel docs pricing](#)
- [AWS Lambda pricing](#) · [Lambda limits](#)
- [AWS MediaConvert pricing](#) · [Mux pricing](#) · [Cloudflare Stream pricing](#) · [Transloadit pricing](#) · [Coconut.co pricing](#)
- [Modal pricing](#) · [RunPod pricing](#) · [Replicate pricing](#) · [Replicate clip-features](#) · [EC2 t3.medium \(Vantage mirror\)](#)

Video intelligence

- [Google Video Intelligence pricing](#) · [AWS Rekognition pricing](#) · [Azure Retail Prices API](#) · [TwelveLabs pricing](#) · [PySceneDetect benchmarks](#)

Vector & data

- [Neon pricing](#) · [Supabase pricing](#) · [Pinecone pricing](#) · [Turbopuffer pricing](#) · [Turbopuffer pricing log](#) · [Qdrant pricing](#) · [Weaviate pricing](#) · [LanceDB](#) · [Upstash pricing](#)

AI models

- [Anthropic pricing docs](#) · [Claude pricing](#) · [Gemini API pricing](#) · [Runway API pricing](#) · [Deepgram pricing](#)

Storage

- [Cloudflare R2 pricing](#) · [AWS S3 pricing](#) · [Backblaze B2 pricing](#) · [Vercel Blob pricing](#)

Auth & analytics

- [Clerk pricing](#) · [PostHog pricing](#) · [Mixpanel pricing](#) · [Plausible](#) · [Dub.co pricing](#) · [Bitly pricing](#)